EXPERIMENT 2

1. AIM

To develop a Python program using Pandas and Matplotlib to read student grade data from a CSV file, validate the data, calculate grade-wise percentages, generate pie and bar charts, identify the highest and lowest grades, and save the charts as image files.

2. ALGORITHM

1. Start the program.
2. Import the Pandas and Matplotlib libraries.
3. Create a CSV file named student_grades.csv with two columns: Grade and Number_of_Students.
4. Store the number of students for grades F, E, D, C, B, A, and S.
5. Read the CSV file using the Pandas library.
6. Validate that all seven grades are present and the total number of students is exactly 40.
7. Calculate the percentage of students for each grade.
8. Display the number and percentage of students for each grade.
9. Generate a pie chart showing the percentage distribution of grades.
10. Generate a bar chart comparing the number of students in each grade.
11. Add suitable titles, labels, legends, and values to both charts.
12. Identify and display the grades with the highest and lowest numbers of students.
13. Save both charts as image files.
14. Stop the program.

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt



# Create CSV file

data = {

   'Grade': ['F', 'E', 'D', 'C', 'B', 'A', 'S'],

   'Number_of_Students': [3, 4, 5, 8, 9, 7, 4]

}
```

```
df = pd.DataFrame(data)
df.to_csv('student_grades.csv', index=False)

# Read CSV file
df = pd.read_csv('student_grades.csv')

# Validate data
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']

if set(df['Grade']) != set(grades) or len(df) != 7:
    print("Invalid grades in CSV file")
elif (df['Number_of_Students'] < 0).any() or not (
    pd.api.types.is_numeric_dtype(df['Number_of_Students'])
):
    print("Invalid student counts")
elif df['Number_of_Students'].sum() != 40:
    print("Total students must be 40")
else:
    # Calculate percentages
    df['Percentage'] = (
        df['Number_of_Students'] / 40 * 100
    )
```

```
print(df.to_string(index=False))


# Pie chart
plt.figure(figsize=(6, 6))
plt.pie(
    df['Number_of_Students'],
    labels=df['Grade'],
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Percentage Distribution of Grades")
plt.legend(df['Grade'], title="Grades")
plt.savefig('grade_pie_chart.png')
plt.show()


# Bar chart
plt.figure(figsize=(7, 5))
bars = plt.bar(
    df['Grade'], df['Number_of_Students'],
    color='skyblue', label='Students'
)
plt.xlabel("Grades")
```

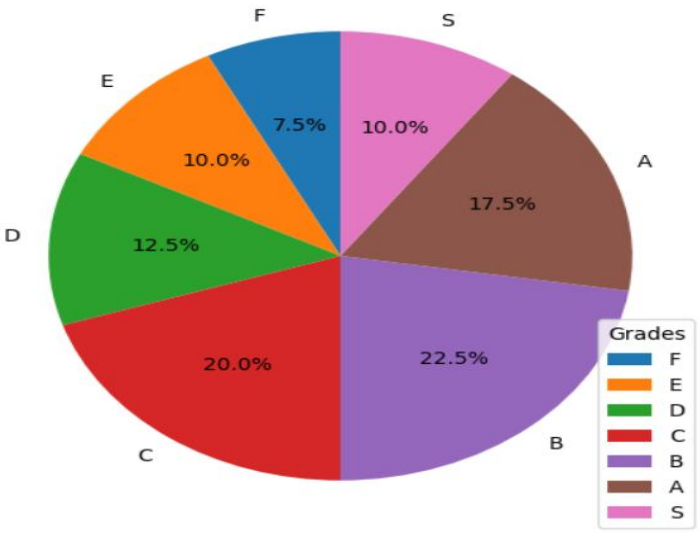
```
plt.ylabel("Number of Students")
plt.title("Grade-wise Student Count")
plt.bar_label(bars)
plt.legend()
plt.savefig('grade_bar_chart.png')
plt.show()

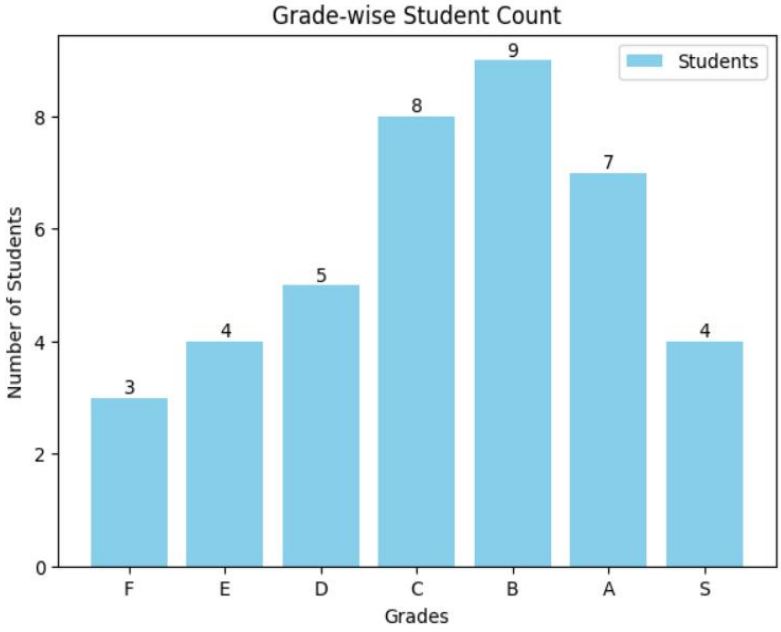
# Highest and lowest grades
max_row = df.loc[df['Number_of_Students'].idxmax()]
min_row = df.loc[df['Number_of_Students'].idxmin()]

print("Highest Grade Count:", max_row['Grade'],
      "-", max_row['Number_of_Students'])
print("Lowest Grade Count:", min_row['Grade'],
      "-", min_row['Number_of_Students'])

print("Charts saved successfully!")
```

OUTPUT:






```
Highest Grade Count: B - 9
Lowest Grade Count: F - 3
Charts saved successfully!
```